Banasthali Vidyapith-Faculty of Mathematics & Computing

MCA III /M.Sc.(DS) III /M.Sc.(CS) I Semester, July-December 2026

Home Assignment (Theory) I

Date: 29-Aug-2026

Course Code: CS 508 **Course Name: Big Data Analytics**

Submission Date: 04 Sep. 2026 **Max Marks: 10**

Q.1 Explain the detailed architecture of MapReduce in Hadoop, including its major components and execution phases. Describe the complete MapReduce workflow with a suitable case study, including a numerical example to calculate the key-value pairs generated and processed at different stages. **(4 Marks)**

Q.2 Discuss the architecture and working of HDFS, highlighting the roles and responsibilities of NameNode, DataNode, Secondary NameNode, and Checkpointing. Illustrate the file storage and retrieval process through a suitable example, along with a numerical problem involving block calculation, block distribution, replication factor, and storage requirements. **(3 Marks)**

Q.3 Describe the architecture and resource-management mechanism of YARN, explaining the roles of ResourceManager, NodeManager, ApplicationMaster, and Containers. Demonstrate the complete lifecycle of a YARN application—from submission and resource allocation to task execution, monitoring, and completion—using a suitable example of resource and container allocation. **(3 Marks)**

Note: Important Instructions for Assignment Submission

- ✓ The assignment must be **handwritten and neatly presented**.
- ✓ A **proper cover page** must be included with the required details.
- ✓ The assignment must be **submitted before/on the given deadline**.
- ✓ Maintain **proper sequence, headings, and numbering** for all questions.